

Assignment No 1

Aim: Implement a robust Java calculator program that captures user input dynamically, processes mathematical expressions using conditional logic and looping constructs, and ensures efficient error handling.

Code:

```
import java.util.Scanner;
public class Calculator {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        int choice = 0;
        int result = 0;

        try {

            do {
                System.out.println("\n--- JAVA CALCULATOR ---");

                System.out.print("Enter first number: ");
                int num1 = sc.nextInt();

                System.out.print("Enter second number: ");
                int num2 = sc.nextInt();

                System.out.println("1.Addition");
                System.out.println("2.Subtraction");
                System.out.println("3.Multiplication");
                System.out.println("4.Division");
                System.out.println("5.Exit");

                System.out.println("Enter your choice : ");
                choice = sc.nextInt();

                switch (choice) {
                    case 1:
                        result = num1 + num2;
                        System.out.println("Addition of " + num1 + " and " + num2 + " : " + result);
                        break;

                    case 2:
                        result = num1 - num2;
                        System.out.println("Subtraction of " + num1 + " and " + num2 + " : " + result);
                        break;

                    case 3:
                        result = num1 * num2;
                        System.out.println("Multiplication of " + num1 + " and " + num2 + " : " + result);
                        break;

                    case 4:
                        result = num1 / num2;
```

```

        System.out.println("Division of " + num1 + " and " + num2 + " : " + result);
        break;

    case 5:
        System.out.println("Exiting program...");
        break;

    default:
        System.out.println("Invalid operator!");
    }

    }while (choice != 5);

    System.out.println("Thank you for using the calculator!");

    }catch (ArithmeticException e) {
        System.out.println("Error: Division by zero is not allowed");
    } catch (Exception e) {
        System.out.println("Error : " + e.getMessage());
    }
}
}

```

Output:

```

--- JAVA CALCULATOR ---
Enter first number: 12
Enter second number: 2
1.Addition
2.Subtraction
3.Multiplication
4.Division
5.Exit
Enter your choice : 1
Addition of 12 and 2 : 14

--- JAVA CALCULATOR ---
Enter first number: 34
Enter second number: 23
1.Addition
2.Subtraction
3.Multiplication
4.Division
5.Exit
Enter your choice : 2
Subtraction of 34 and 23 : 11

```

```
--- JAVA CALCULATOR ---  
Enter first number: 4  
Enter second number: 5  
1.Addition  
2.Subtraction  
3.Multiplication  
4.Division  
5.Exit  
Enter your choice : 3  
Multiplication of 4 and 5 : 20
```

```
--- JAVA CALCULATOR ---  
Enter first number: 12  
Enter second number: 2  
1.Addition  
2.Subtraction  
3.Multiplication  
4.Division  
5.Exit  
Enter your choice : 4  
Division of 12 and 2 : 6
```

```
--- JAVA CALCULATOR ---  
Enter first number: 5  
Enter second number: 2  
1.Addition  
2.Subtraction  
3.Multiplication  
4.Division  
5.Exit  
Enter your choice : 5  
Exiting program...  
Thank you for using the calculator!
```